

19.5 Random regressors and a curved mean

When the regressors are random, as in almost every observational study, the linear model is read conditionally on $\mathbf{X}$. [Section 14.1](#) settled the case in which the conditional model is right. If $\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, every t and F procedure keeps its exact level unconditionally, whatever the distribution of the regressors (Theorem 14.1.2), and $\hat{\boldsymbol{\beta}}$ is unbiased with covariance $\sigma^2 \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1}]$ (Corollary 5.5.5). Nothing further needs to be proved about that case.

Random regressors create new problems only when the conditional model fails, and it can fail in two essentially different ways. The regressors can be correlated with the errors, as when they are measured with error or when a common cause of regressor and response is omitted. Then least squares does not even estimate a coefficient of interest, a problem taken up in [Chapter 24](#) and [Chapter 25](#). Or the conditional mean $\mathbb{E}(Y \mid \mathbf{x})$ is not linear in $\mathbf{x}$. This section treats the second case. Its message is that a curved mean, fitted by a linear model with random regressors, behaves like heteroscedastic error, and the usual standard errors are wrong even when the true errors are homoscedastic.

19.5.1 What least squares estimates, conditionally

Proposition 19.5.1 — The design-dependent target

Let $(\mathbf{X}, \mathbf{Y})$ be random with $\text{rank}(\mathbf{X}) = p < n$ almost surely, $\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) = \boldsymbol{\mu}(\mathbf{X})$ for some function $\boldsymbol{\mu}$ with values in $\mathbb{R}^n$, not necessarily in $\mathcal{C}(\mathbf{X})$, and $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$. Put $\boldsymbol{\beta}_{\mathbf{X}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu}(\mathbf{X})$.

- (a) $\mathbb{E}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta}_{\mathbf{X}}$ and $\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$.
- (b) If the second moments below are finite, $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}_{\mathbf{X}})$.
- (c) $\mathbb{E}(s^2 \mid \mathbf{X}) = \sigma^2 + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}(\mathbf{X})\|^2 / (n - p)$.

Proof.

- (a) Given $\mathbf{X}$, apply Proposition 5.5.7 with $\boldsymbol{\delta} = \boldsymbol{\mu}(\mathbf{X}) - \mathbf{X}\boldsymbol{\beta}_{\mathbf{X}}$, which satisfies $\mathbf{X}^\top \boldsymbol{\delta} = \mathbf{0}$, and Theorem 5.5.1(b). (b) is the law of total covariance (Proposition 2.5.4) applied to (a). (c) is Eq. 2.3.1 given $\mathbf{X}$.

■

Given the design, $\hat{\boldsymbol{\beta}}$ scatters around $\boldsymbol{\beta}_{\mathbf{X}}$ exactly as the usual formula says. But $\boldsymbol{\beta}_{\mathbf{X}}$, the least squares coefficient of the true mean over the design drawn, changes from design to design. When $\boldsymbol{\mu}(\mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ it is constant (Corollary 5.5.5). Otherwise (b) adds a *between-design* component that no conditional calculation can see, while (c) inflates s^2 by the misfit, which compensates only partly.

19.5.2 The large-sample picture

The natural fixed target is the population projection of [Section 6.11](#), $\boldsymbol{\beta}^* = \boldsymbol{\Sigma}^{-1} \mathbb{E}(\mathbf{x}Y)$, which least squares estimates consistently whatever the form of the mean (Proposition 6.11.3). Its

sampling distribution is the following.

Theorem 19.5.2 — Least squares for the population projection

Let $(\mathbf{x}_i, Y_i)$, $i = 1, \dots, n$, be independent copies of $(\mathbf{x}, Y)$, where $\mathbf{x} \in \mathbb{R}^p$ (whose first component may be the constant 1) and Y have finite fourth moments and $\Sigma = E(\mathbf{x}\mathbf{x}^\top)$ is positive definite. Let $\beta^* = \Sigma^{-1} E(\mathbf{x}Y)$, $U = Y - \mathbf{x}^\top \beta^*$ and $\Omega = E(U^2 \mathbf{x}\mathbf{x}^\top)$. Then $E(\mathbf{x}U) = \mathbf{0}$, and as $n \rightarrow \infty$:

- (a) $\sqrt{n}(\hat{\beta} - \beta^*) \rightarrow N_p(\mathbf{0}, \Sigma^{-1}\Omega\Sigma^{-1})$ in distribution;
- (b) $s^2 \rightarrow E(U^2)$ and $n s^2(\mathbf{X}^\top \mathbf{X})^{-1} \rightarrow E(U^2) \Sigma^{-1}$ in probability;
- (c) if $E(U^2 | \mathbf{x}) = E(U^2)$ almost surely, the two limiting covariances agree. Otherwise the usual interval for $\mathbf{a}^\top \beta^*$ of nominal level $1 - \alpha$ has coverage tending to $2\Phi(z_{\alpha/2}\sqrt{\rho_{\mathbf{a}}}) - 1$, with $\rho_{\mathbf{a}} = E(U^2) \mathbf{a}^\top \Sigma^{-1} \mathbf{a} / \mathbf{a}^\top \Sigma^{-1} \Omega \Sigma^{-1} \mathbf{a}$.

Proof. $E(\mathbf{x}U) = E(\mathbf{x}Y) - \Sigma\beta^* = \mathbf{0}$. Write $\mathbf{U} = \mathbf{Y} - \mathbf{X}\beta^*$, so that

$$\sqrt{n}(\hat{\beta} - \beta^*) = \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} \right)^{-1} \frac{\mathbf{X}^\top \mathbf{U}}{\sqrt{n}}.$$

By the weak law of large numbers $\mathbf{X}^\top \mathbf{X}/n = n^{-1} \sum_i \mathbf{x}_i \mathbf{x}_i^\top \rightarrow \Sigma$ in probability, so it is invertible with probability tending to one and its inverse tends to Σ^{-1} , inversion being continuous at Σ . The vectors $\mathbf{x}_i U_i$ are independent with mean $\mathbf{0}$ and covariance Ω , which is finite because $|E(U^2 x_j x_k)| \leq (E U^4 E x_j^2 x_k^2)^{1/2}$ and U has a finite fourth moment. By the central limit theorem $\mathbf{X}^\top \mathbf{U}/\sqrt{n} = \sqrt{n} \cdot n^{-1} \sum_i \mathbf{x}_i U_i \rightarrow N_p(\mathbf{0}, \Omega)$. If a random matrix $\mathbf{A}_n$ tends in probability to a constant $\mathbf{A}$ and a random vector $\mathbf{T}_n$ tends in distribution to $\mathbf{T}$, then $\mathbf{A}_n \mathbf{T}_n \rightarrow \mathbf{A} \mathbf{T}$: the difference $(\mathbf{A}_n - \mathbf{A}) \mathbf{T}_n$ tends to zero in probability because $\mathbf{T}_n$ is bounded in probability, $\mathbf{A} \mathbf{T}_n \rightarrow \mathbf{A} \mathbf{T}$ by continuous mapping, and Slutsky's lemma combines the two. This proves (a).

- (b) $(n-p)s^2 = \mathbf{U}^\top (\mathbf{I} - \mathbf{M}) \mathbf{U} = \mathbf{U}^\top \mathbf{U} - Q_n$, where $Q_n = (\mathbf{X}^\top \mathbf{U}/\sqrt{n})^\top (\mathbf{X}^\top \mathbf{X}/n)^{-1} (\mathbf{X}^\top \mathbf{U}/\sqrt{n})$ converges in distribution by the argument for (a) and continuous mapping. Hence $Q_n/n \rightarrow 0$ in probability, while $\mathbf{U}^\top \mathbf{U}/n \rightarrow E(U^2)$ by the law of large numbers. Then $n s^2 (\mathbf{X}^\top \mathbf{X})^{-1} = s^2 (\mathbf{X}^\top \mathbf{X}/n)^{-1} \rightarrow E(U^2) \Sigma^{-1}$.
- (c) If $E(U^2 | \mathbf{x}) = c$, then $\Omega = E[E(U^2 | \mathbf{x}) \mathbf{x}\mathbf{x}^\top] = c \Sigma$, with $c = E(U^2)$, and $\Sigma^{-1} \Omega \Sigma^{-1} = c \Sigma^{-1}$. In general, the t statistic for $\mathbf{a}^\top \beta^*$ is $\sqrt{n} \mathbf{a}^\top (\hat{\beta} - \beta^*)$ divided by $(n s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a})^{1/2}$. By (a), (b) and Slutsky's lemma it tends to $N(0, 1/\rho_{\mathbf{a}})$, and the coverage follows as in Theorem 19.3.2(c).

■

The condition in (c) holds when the mean is linear and the errors are homoscedastic. It typically fails when either assumption fails ([Exercise B2](#)) gives an exception), and a curved mean alone is enough. If $E(Y | \mathbf{x}) = m(\mathbf{x})$ and $\text{Var}(Y | \mathbf{x}) = \sigma^2$, then $U = (m(\mathbf{x}) - \mathbf{x}^\top \beta^*) + \varepsilon$ and

$$E(U^2 | \mathbf{x}) = (m(\mathbf{x}) - \mathbf{x}^\top \beta^*)^2 + \sigma^2,$$

which varies with $\mathbf{x}$ wherever the linear approximation is poor, exactly like a heteroscedastic error. The correct covariance $\Sigma^{-1} \Omega \Sigma^{-1}$ is a population version of the sandwich Eq. 5.5.3, and [Chapter 21](#) shows how to estimate it.

Example. (A line through a parabola)

Let X be uniform on $(0, 2)$, $E(Y | X) = X^2$ and $\text{Var}(Y | X) = 0.04$, with normal errors. A straight line is fitted. The population line has slope $\beta^* = \text{Cov}(X, X^2) / \text{Var}(X) = 2$ and intercept $-2/3$, and the approximation error is $U_0^2 - 1/3$ with $U_0 = X - 1$ uniform on $(-1, 1)$. The slope's component of $\Sigma^{-1}\Omega\Sigma^{-1}$ is $E[(X - 1)^2 U^2] / \text{Var}(X)^2$ (Exercise A1), which equals $9(44/945 + 0.04/3) = 0.5390$. The usual formula estimates $E(U^2) / \text{Var}(X) = 3(4/45 + 0.04) = 0.3867$ instead. The true standard error is 1.181 times what the usual one estimates, and by Theorem 19.5.2(c) the nominal 95% interval for β^* has limiting coverage 0.903.

Simulation agrees. With 40000 samples the interval covered $\beta^* = 2$ in 0.896 of samples of size 50 and 0.900 of size 400; with 10000 samples of size 1600, in 0.908. Over the same samples, n times the variance of the slope was 0.584, 0.546 and 0.530, approaching the sandwich value 0.5390, not the usual 0.3867. As a statement about the design-dependent target β_X , the same intervals are *conservative*. They covered it in 0.999 of the samples of size 50, because s^2 estimates $E(U^2) = 0.1289$ rather than $\sigma^2 = 0.04$. In a larger simulation of 200000 samples of size 50, the variance of the slope, 0.01163, splits as in Proposition 19.5.1(b) into 0.00249 from the errors and 0.00913 from the variation of β_X between designs.

```
import numpy as np
from scipy import stats
from fractions import Fraction as Fr
sigma2 = Fr(1, 25)                                # sigma = 0.2
EU2, EU4, EU6 = Fr(1, 3), Fr(1, 5), Fr(1, 7)        # moments of U = X - 1 ~
    Uniform(-1, 1)
Eu2 = EU4 - Fr(2, 3) * EU2 + Fr(1, 9) + sigma2      # u = U^2 - 1/3 + error
sandwich = (EU6 - Fr(2, 3) * EU4 + Fr(1, 9) * EU2 + sigma2 * EU2) / EU2**2
usual = Eu2 / EU2
print("n Var(slope): sandwich", float(sandwich), " usual", float(usual))
z = stats.norm.ppf(0.975)
print("limiting coverage of the usual interval:",
      2 * stats.norm.cdf(z * np.sqrt(float(usual / sandwich))) - 1)
```

[Run this code online](#)

```
def one_run(n, reps, rng):
    x = rng.uniform(0, 2, size=(reps, n))
    y = x**2 + 0.2 * rng.normal(size=(reps, n))
    xc = x - x.mean(axis=1, keepdims=True)
    Sxx = np.sum(xc**2, axis=1)
    b = np.sum(xc * y, axis=1) / Sxx                # least squares slope
    b_X = np.sum(xc * x**2, axis=1) / Sxx           # its conditional mean
    given X
    a = y.mean(axis=1) - b * x.mean(axis=1)
    s2 = np.sum((y - a[:, None] - b[:, None] * x) ** 2, axis=1) / (n - 2)
    half = stats.t.ppf(0.975, n - 2) * np.sqrt(s2 / Sxx)
    return np.mean(np.abs(b - 2.0) <= half), np.mean(np.abs(b - b_X) <= half), n * b
    .var()

rng = np.random.default_rng(1905)
for n in [50, 400]:
    c_star, c_X, nvar = one_run(n, 4000, rng)
```

```
print(f"n = {n}: covers beta* {c_star:.3f}, covers beta_X {c_X:.3f}, n Var(slope) {nvar:.3f}")
```

[Run this code online](#)

Which target is right depends on the question: β^* summarizes the population, $\beta_{\mathbf{X}}$ this design. With a curved mean, neither describes $E(Y \mid \mathbf{x})$ itself.

19.5.3 Controlled regressors

One situation with random regressors is more benign than it looks. In a controlled experiment the analyst sets a nominal value x_i , but the value actually delivered is $x_i + \delta_i$, with an error δ_i independent of x_i . If the response follows $\beta_0 + \beta_1(x_i + \delta_i) + \varepsilon_i$, then regressing on the nominal values gives

$$Y_i = \beta_0 + \beta_1 x_i + (\beta_1 \delta_i + \varepsilon_i),$$

a correct linear model with a larger error variance ([Exercise \(B1\)](#)). Least squares on the nominal settings is unbiased, as Berkson (1950) pointed out. Contrast the *classical* measurement error model, in which the true value is fixed and the *recorded* value carries the error. There the error is correlated with the recorded regressor, and least squares is biased towards zero (Theorem 24.1.1).

Rounding can fall on either side. If the grid value is what was set, as with a dial, the model is Berkson's. If a continuous true value is *recorded* rounded to a grid of width w , the rounding error is roughly uniform with variance $w^2/12$ and roughly independent of the true value, so the error is classical. In simple regression, classical error of variance τ^2 multiplies the slope by $\text{Var}(x)/(\text{Var}(x) + \tau^2)$ ([Chapter 24](#)). So measurement error, rounding included, can be ignored when its variance is a small fraction of $\text{Var}(x)$.

Idea. With random regressors, what matters is whether the conditional model holds. If it does, nothing changes; if the mean is curved, $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is not the covariance.

19.5.4 Exercises

19.5.4.1 A. Check your understanding

Exercise A1.

With $\mathbf{x} = (1, X)^\top$, show that the slope entries of $\Sigma^{-1} \Omega \Sigma^{-1}$ and $E(U^2) \Sigma^{-1}$ are $E[(X - \mu_X)^2 U^2] / \text{Var}(X)^2$ and $E(U^2) / \text{Var}(X)$.

Solution. $\Sigma = \begin{pmatrix} 1 & \mu_X \\ \mu_X & E X^2 \end{pmatrix}$ has determinant $\text{Var}(X)$, and the second row of Σ^{-1} is $(-\mu_X, 1) / \text{Var}(X)$. So $\mathbf{e}_2^\top \Sigma^{-1} \mathbf{x} = (X - \mu_X) / \text{Var}(X)$, and $\mathbf{e}_2^\top \Sigma^{-1} \Omega \Sigma^{-1} \mathbf{e}_2 = E[(X - \mu_X)^2 U^2] / \text{Var}(X)^2$. Similarly $[\Sigma^{-1}]_{22} = 1 / \text{Var}(X)$.

Exercise A2.

In [Proposition 19.5.1](#), show that $\text{Cov}(\beta_{\mathbf{X}}) = \mathbf{0}$ when $\mu(\mathbf{X}) = \mathbf{X}\beta$, and give an example

of a nonlinear μ for which $\text{Cov}(\beta_{\mathbf{X}}) = \mathbf{0}$ anyway.

19.5.4.2 B. Practice

Exercise B1.

In the controlled-regressor model, let δ_i and ε_i be independent, with variances τ^2 and σ^2 , independent of the settings. Show that least squares on the nominal x_i is unbiased for (β_0, β_1) , that the usual standard errors are valid with error variance $\sigma^2 + \beta_1^2 \tau^2$, and that s^2 estimates this variance.

Solution. The composite errors $\eta_i = \beta_1 \delta_i + \varepsilon_i$ are independent of the fixed settings, with mean 0 and variance $\sigma^2 + \beta_1^2 \tau^2$, and uncorrelated across i . The model $Y_i = \beta_0 + \beta_1 x_i + \eta_i$ therefore satisfies the second-moment assumptions of [Chapter 5](#), so Theorem 5.5.1 and Theorem 5.6.1 apply with σ^2 replaced by $\sigma^2 + \beta_1^2 \tau^2$. If δ_i and ε_i are normal, so is η_i , and the exact theory of Part III applies too.

Exercise B2.

In [Theorem 19.5.2](#), show that $\Sigma^{-1} \Omega \Sigma^{-1} = E(U^2) \Sigma^{-1}$ iff $E[(E(U^2 | \mathbf{x}) - E(U^2)) \mathbf{x} \mathbf{x}^\top] = \mathbf{0}$. Give an example of a nonconstant $E(U^2 | \mathbf{x})$ for which the usual standard errors are nevertheless consistent.

19.5.4.3 C. Going deeper

Exercise C1.

In [Theorem 19.5.2](#), let $E(Y | \mathbf{x}) = m(\mathbf{x})$, $\text{Var}(Y | \mathbf{x}) = \sigma^2$, and put $d(\mathbf{x}) = m(\mathbf{x}) - \mathbf{x}^\top \beta^*$. Show that $\sqrt{n}(\beta_{\mathbf{X}} - \beta^*) \rightarrow N_p(\mathbf{0}, \Sigma^{-1} E(d^2 \mathbf{x} \mathbf{x}^\top) \Sigma^{-1})$, and that $\Sigma^{-1} \Omega \Sigma^{-1} = \sigma^2 \Sigma^{-1} + \Sigma^{-1} E(d^2 \mathbf{x} \mathbf{x}^\top) \Sigma^{-1}$. Interpret the two terms in the light of [Proposition 19.5.1\(b\)](#).

Solution. $\beta_{\mathbf{X}} - \beta^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{d}$ with $\mathbf{d} = (d(\mathbf{x}_1), \dots, d(\mathbf{x}_n))^\top$, and $E(\mathbf{x} d(\mathbf{x})) = E(\mathbf{x} U) = \mathbf{0}$ because $E(U | \mathbf{x}) = d(\mathbf{x})$. The argument of [Theorem 19.5.2\(a\)](#) with U replaced by $d(\mathbf{x})$ gives the first limit. Since $U = d(\mathbf{x}) + \varepsilon$ with $E(\varepsilon | \mathbf{x}) = 0$, $E(U^2 | \mathbf{x}) = d(\mathbf{x})^2 + \sigma^2$, and $\Omega = \sigma^2 \Sigma + E(d^2 \mathbf{x} \mathbf{x}^\top)$. The first term, $\sigma^2 \Sigma^{-1}$, is the asymptotic version of $n \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$, the within-design part; the second is the asymptotic version of $n \text{Cov}(\beta_{\mathbf{X}})$, the between-design part. (They are limits of those moments only under uniform integrability, and for some designs the finite-sample moments do not even exist.)